Status	Finished
Started	Sunday, 25 October 2026, 12:28 PM
Completed	Sunday, 25 October 2026, 12:32 PM
Duration	3 mins 32 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O👎
c. O(log n) ✓
d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search continues in the left sublist
c. The list is sorted
d. The search ends successfully ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list contains duplicate elements
b. When the list is sorted ✓
c. When the list is unsorted
d. When the list is very small

Question **4**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The middle element is chosen randomly
b. The search stops
c. The lower middle element is chosen as the middle element
d. The higher middle element is chosen as the middle element ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search backtracks to find duplicate elements
d. The search continues until the end of the list

Question **6**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓
b. Inserting elements into a list
c. Sorting elements in a list
d. Deleting elements from a list

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the middle element in the list ✓
c. Compare the target element with the first element in the list
d. Sort the list

Question **8**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O👎
c. O(log n)
d. O(1) ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It can only be applied to large lists

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000
b. 100
c. 500
d. 10 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the left sublist
d. In the middle of the list

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. There must be a mechanism to access middle element directly
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **14**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Hash
c. Both (1) & (3)
d. Linear

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O👎 ✓
b. O(1)
c. O(n log n)
d. O(log n)